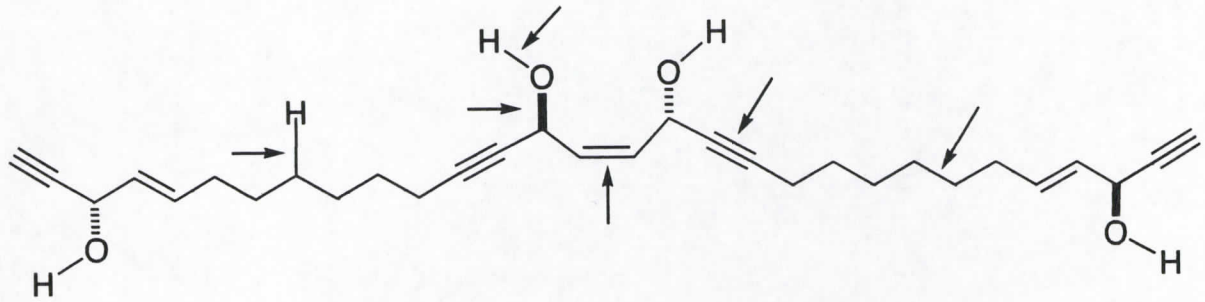
Organic Division Cumulative Exam October 2 , 2015

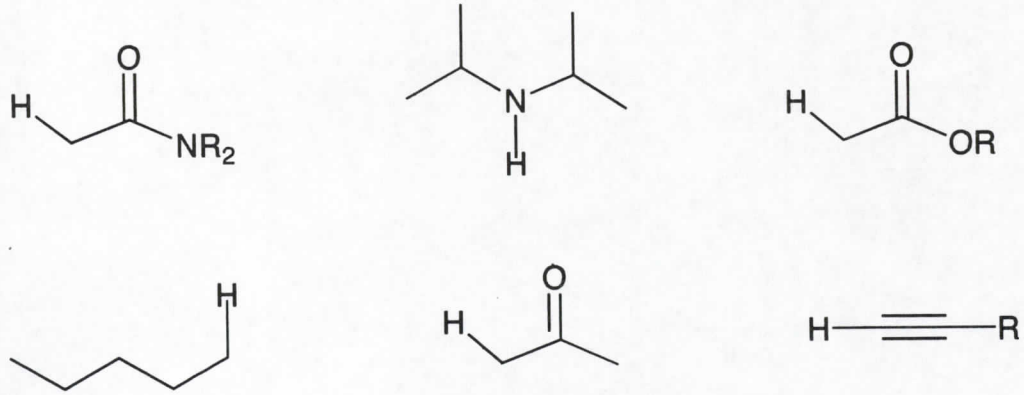
Topic: Important numbers in organic chemistry. There are not a lot of numbers you need to know in organic chemistry, but this cume will test your knowledge some of the more important ones, including bond lengths, strengths, pK_a's and A-values. **If you don't know these numbers, you should familiarize yourself with them.**

1. Give the bond lengths to within 0.1 Å for an **average** bond of the indicated type.
2. Give the bond strengths (homolytic bond dissociation energies) rounded to the nearest 5 kcal/mol. For the multiple bonds, give the **total** bond energy (i.e. single + double, or single + double + triple).



A cytotoxic polyacetylene isolated from the marine sponge *Petrosia*

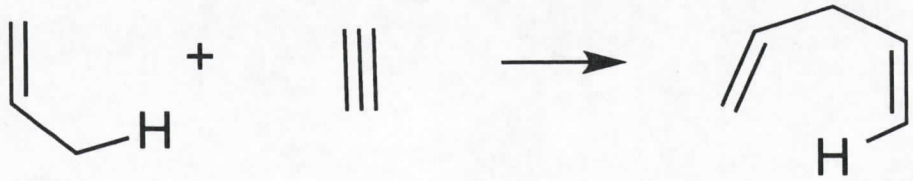
3. Give the approximate pK_a's of the following compounds.



4. Give A-values in kcal/mol to within 0.2 kcal/mol (A-value = free energy change for the conversion of the equatorial to axial substituent in cyclohexane).

R	A value
≡—H	
$C(CH_3)_3$	
OCH_3	
CH_3	
phenyl	
CH_2CH_3	
$CH{=}CH_2$	

5. Estimate ΔH for the following reactions. Show how you arrived at your answers.

(a)

(b)